What is meant by Transposition of the line conductors?

Transposition of line conductors means changing the position of the three phases on the line supports twice over the total length of the line. The line conductors in practice are so transposed that each of the three possible arrangements of conductors exist for one third of the total length of the line.



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If the frequency of the transmission line increases then:

a) Line resistance increases

b) Line resistance decreases

c) Shunt rectance increases

d) Shunt reactance decreases

Answer (d). If the frequency of the line increases, then shunt reactance = 1/(2 x pi x f x c). So when frequency (f) inceases shunt reactance decreases.

The phenomenon in which the conductor surface carries more current compared to core when alternative voltage is applied is



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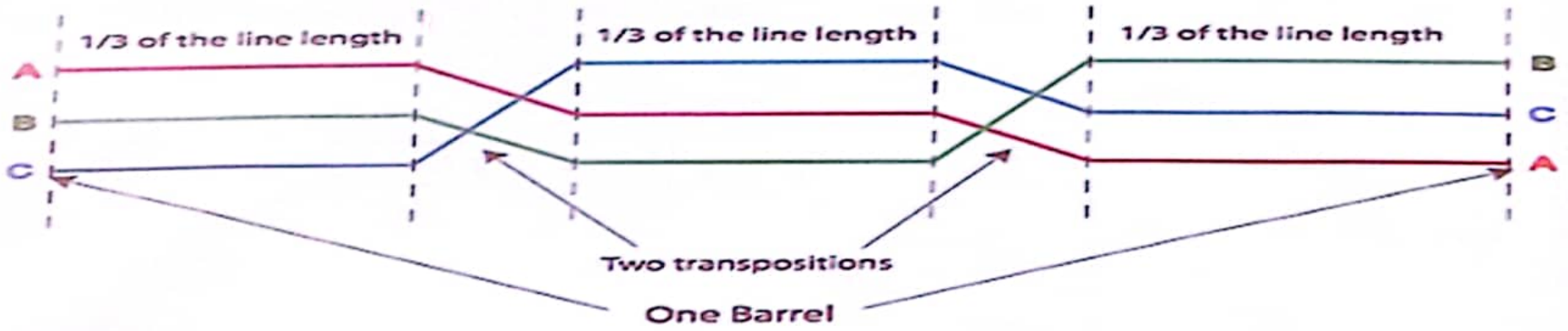


Figure 5 – Barrel

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a) Corona

b) Skin Effect

c) Ferranti Effect

d) Lenz's Law

Answer (a). This phenomeon is called Skin Effect



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In overhead lines for transmitting power we generally use:

a) Copper Conductors

b) Aluminum Conductors

c) ACSR conductors

d) Galvanized Steel Conductors

ACSR Conductors implies:

a) Aluminum Conductors Steel Reinforced

b) Anodised Core Steel Reinforced

c) All Conductors Surface Reinforced

d) None of These

ACSR Conductors are used in overhead transmission lines in place of copper because:



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The function of steel wire in the ACSR conductors is to:

a) Compensate for Skin Effect

b) Provide additional mechanical strenght

c) Carry large currents

d) To Reduce Inductance

Which material is used for manufacturing of ground wire:

a) Aluminum

b) Steel wire

c) Cast iron

d) Galvanised Steel

Answer (b). Galvanised Steel in the ACSR conductors provide better mechanical strenght while the aluminum strands on the outer diameter used to carry the bulk current.

Answer (d). Galvanised Steel wire is cheap in cost and withstand corrosion, so this is used as ground wire.

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c) All Conductors Surface Reinforced

d) None of These

ACSR Conductors are used in overhead transmission lines in place of copper because:

a) Economy factor

b) They are light weight

c) All of these

d) High Tensile Strenght

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How transmission voltage is selected

Transmission voltage is stepped up from power plant in order to reduce the losses in transmission. But for every transmission line there is a particular fixed limit for voltage to be transmitted. Beyond this limit the rise in voltage will not result in any economical advantage. Cost analysis is required for finding the optimum voltage; this analysis includes calculating the cost of the conductors, cost of insulators, supports, transformers switchgear, lightning arresters and cost of erection. All the tota cost should be minimum for a particular voltage.

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Which is the top most conductor in the transmission:

a) R-Phase Conductor

b) S-Phase Conductor

c) T-Phase Conductor

d) Earth Conductor



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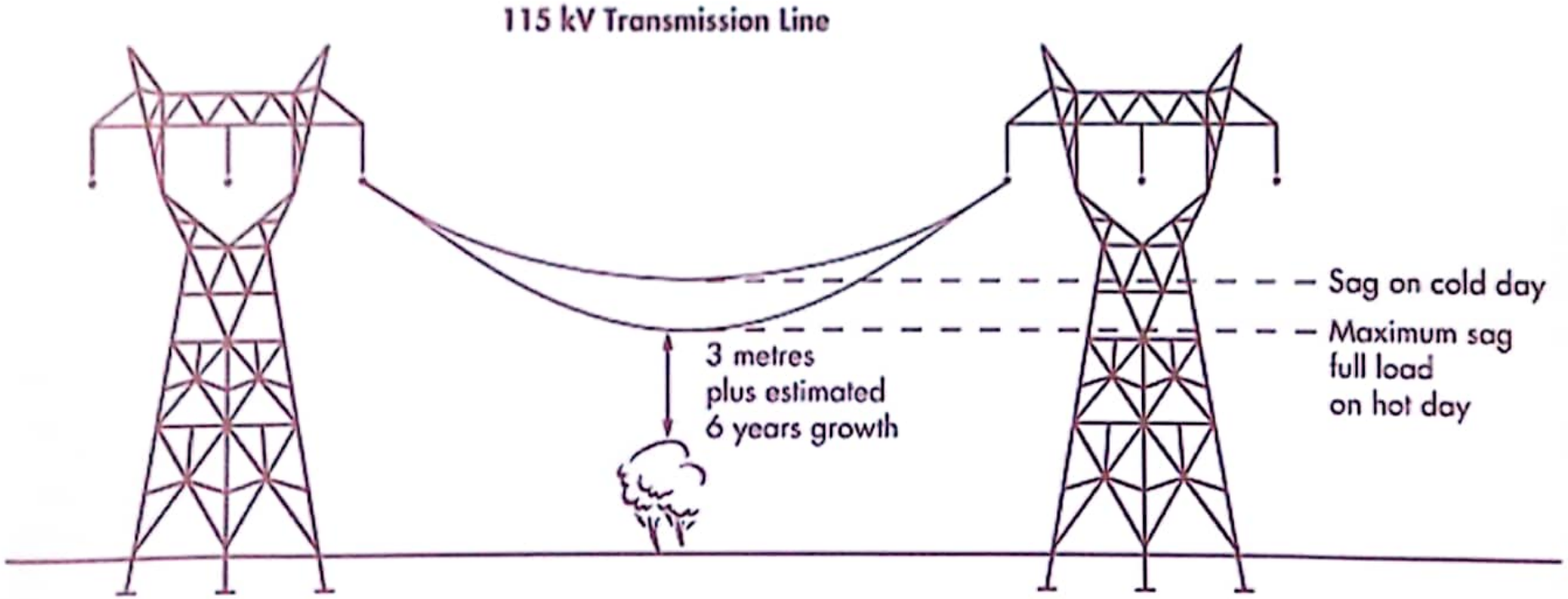
Effect of Temperature rise in overhead lines is to:

a) Increases the SAG and decreases the Tension of lines

b) Decreases the SAG and Increases the Tension of lines

c) Increase in both SAG and Tension of the lines

d) Decreases in both the SAG and Tension of the lines



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List the advantages of per unit computations.

(1) The per unit impedance referred to either side of a single phase transformer is the same

(2) The per unit impedance referred to either side of a single phase transformer is the same.

(3) The per unit impedance referred to either side of a three phase transformer is the same regardless of the three phase connections whether they are Y-Y, Δ-Δ or Δ-Y

(4) The chance of confusion between the line and phase quantities in a three phase balanced system is greatly reduced.

(5) The manufacturers usually provide the impedance values in per unit

(6) The computational effort in power system is very much reduced with the use of per unit quantities.

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Araçlar İmzala Yorum

What is transient state of the power system?

The state of the system in the first second following a system fault or large disturbance is called the transient state of the power system.

What is the need for base values?

The components of various sections of power system may operate at different voltage and power levels. It will be convenient for analysis of power system if the voltage, power, current and impedance ratings of power system components are expressed with reference to a common value called base value. Then all the voltages, power, current and impedance ratings of the components are expressed as a

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Why faults occur in a power system?

Faults occur in a power system due to insulation failure of equipment's, flashover of lines initiated by a lightening stroke, permanent damage to conductors and towers or accidental faulty operations.

How are the faults classified? In one method, the faults are classified as,

Symmetrical faults - fault currents are equal in all the phases and can be analyzed on per phase basis

Unsymmetrical faults – fault currents are unbalanced and so they can be analyzed only using symmetrical componenets

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Which sequence network contains a voltage source

a)Negative b)zero c)Equal d)positive

The reference bus for positive sequence network is system

a)Neutral b)negative c)positive d)zero

The total number of sequence voltage components in a three phase system is

a)10 b)9 c)5 d)6

What is the need for short circuit studies or fault analysis?

The short circuit studies are essential in order to design or develop the protective schemes for various parts of the system. The protective scheme consists of current and voltage sensing devices, protective relays and circuit breakers. The selection of these devices mainly depends on various currents that may flow in the fault conditions.

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Araçlar İmzala Yorum

What are the factors that affect the transient stability?

The transient stability is generally affected by two factors namely, (1) Type of fault (2) Location of fault.

What is meant by stability study?

The procedure of determining the stability of a system upon occurrence of a disturbance followed by various switch off and switch on actions is called a stability study.

What is meant by short circuit fault?

Short circuit faults involve power conductor or conductors-to-ground or short circuit between conductors. These faults are characterized by increase in current and fall in voltage and frequency.

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What are symmetrical components?

An unbalanced system of N related vectors can be resolved into N systems of balanced vectors called symmetrical components.

Positive sequence components
Negative sequence components
Zero sequence components

In which fault zero sequence network is not connected

a)line to ground
b)line to line
c)ground to neutral
d)line to neutral

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What is meant by symmetrical fault?

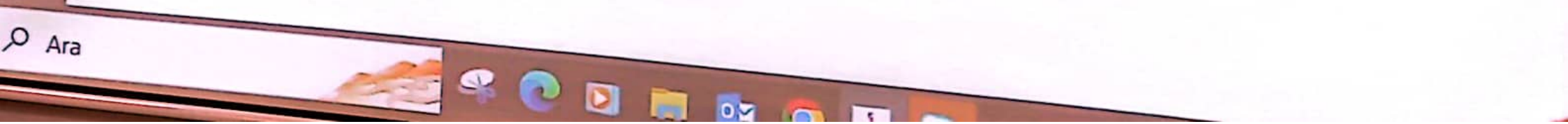
The fault is called symmetrical fault if the fault current is equal in all the phases. This fault conditions are analyzed on per phase basis using Thevenin's theorem or using bus impedance matrix. The three-phase fault is the only symmetrical fault.

The real power depends on

a)power factor b)power angle c)power reactive d)voltage

The most common buses in a power system are

a)slack bus b)generator bus c)load bus d)A&B



CamScanner ile tarandı

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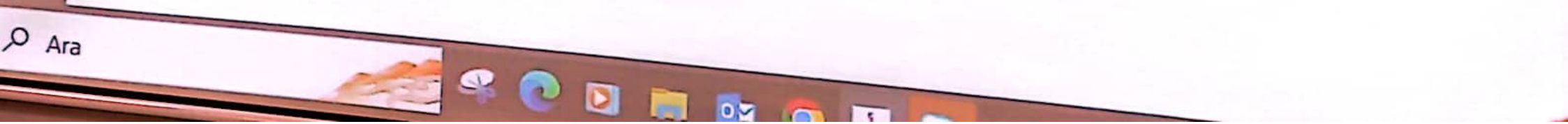
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Fault Types

- There are two main types of faults
  - symmetric faults: system remains balanced; these faults are relatively rare, but are the easiest to analyze so we'll consider them first.
  - unsymmetric faults: system is no longer balanced; very common, but more difficult to analyze
- The most common type of fault on a three phase system by far is the single line-to-ground (SLG), followed by the line-to-line faults (LL), double line-to-ground (DLG) faults, and balanced three phase faults

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Bundled conductors used in transmission to reduces the

a) Surge impedance of line
b) Voltage drop of the line
c) corona loss of the line
d) I^2R loss of the line



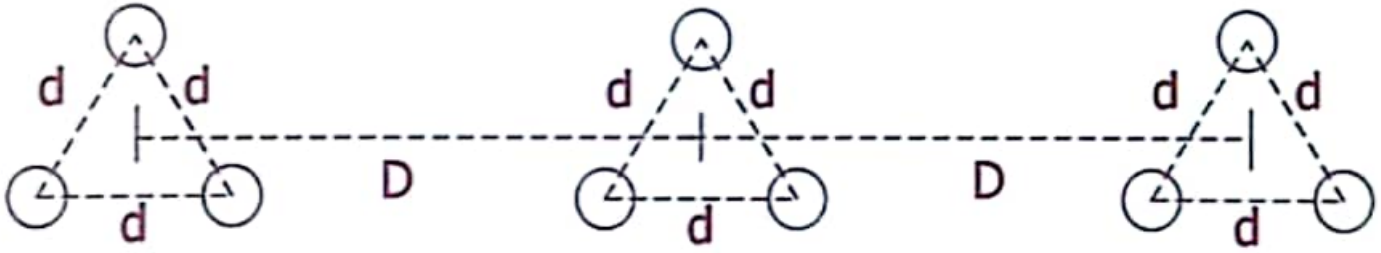


Fig-E: GMD of Bundled conductors. The bundled conductors are placed horizontally. The distance between two extreme conductors is '2D'

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[6] Power transformers are generally designed to have maximum efficiency around

A. No load

B. Half load

C. Near full load

D. 10% overload

[7] The transformer laminations are insulated from each other by

A. Mica strip

B. Thin coat of varnish

C. Paper

D. Any of the above

[8] A transformer core is laminated to

A. Reduce hysteresis loss

B. Reduce eddy current losses

C. Reduce copper losses

D. Reduce all above losses

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For power-system fault studies, it is assumed that the system is operating under balanced steady-state conditions prior to the fault, and sequence networks are uncoupled before the fault occurs.
(a) True (b) False

The first step in power-system fault calculations is to develop sequence networks based on the single-line diagram of the system, and then reduce them to their Thevenin equivalents, as viewed from the fault location.
(a) True (b) False

When calculating symmetrical three-phase fault currents, only _____ sequence network needs to be considered.

In order of frequency of occurance of short-circuit faults in three-phase power systems, list those: _____, _____, _____, _____.

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